

Stockd.

Project Proposal

Computer Vision

AI-Powered Smart Fridge Retrofit System

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Know what
you have.
Waste less.

Introduction

Stockd. is a smart-fridge retrofit system that automatically tracks food as it enters and leaves a refrigerator. A camera and Raspberry Pi use computer vision and AI to identify items and maintain a digital inventory.

Then, through the Stockd. mobile app, users are able to see their kitchen inventory, get recommendations on what to eat by when, along with active tracking of real or estimated expiration dates and quantities per item. Users are also able to add items from other parts of their kitchen such as a pantry or cabinet and also update/change any inventory manually if necessary.



A smarter
fridge for
every home

The Problem

Food gets forgotten.



Food expires
before it gets used



People buy
duplicate groceries



It's hard to know
what to cook



Manual tracking is
time-consuming

People often forget what they already have, food expires before it gets used, and meal planning starts without an accurate picture of what is in the fridge.



1/3

of all food produced
globally is wasted

-UNEP (United Nations
Environment Programme)

Our Solution

Turn any refrigerator into a smart refrigerator.

Stockd. uses a camera and Raspberry Pi to automatically detect food entering and leaving the fridge, maintain a real-time inventory, track freshness, and recommend meals based on what the user already has.



**Automatically
detects food
items**



**Track items entering
and leaving**

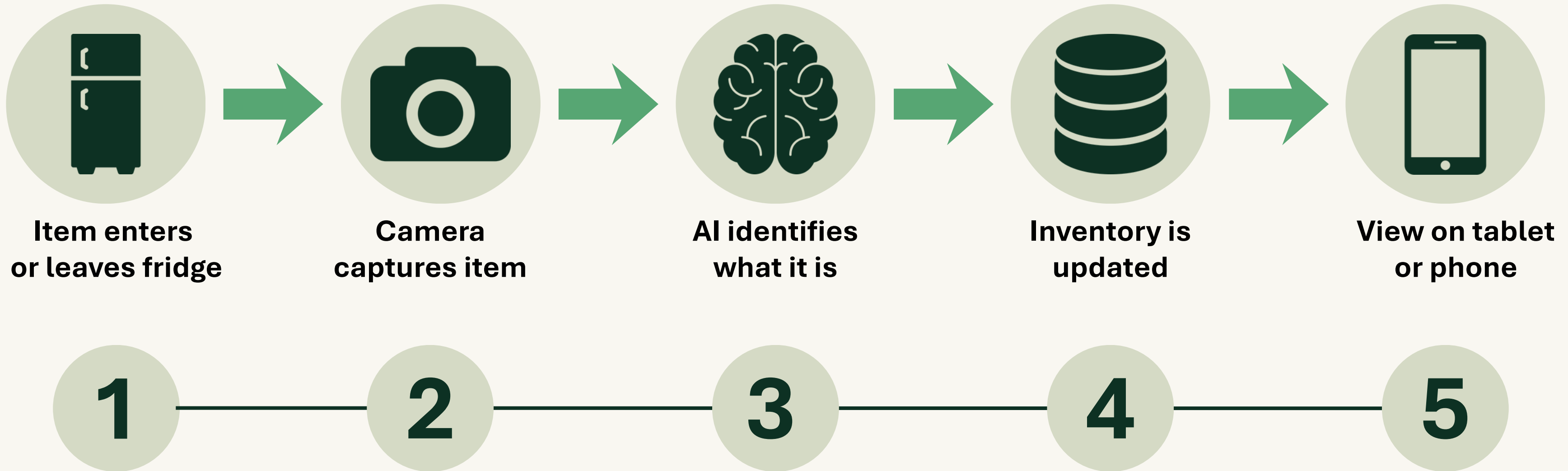


**Monitors freshness
and recommends meals**



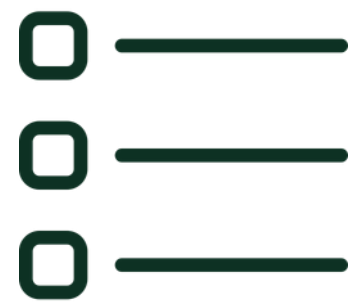
How It Works

From physical food to a digital inventory.



Key Features

More than just inventory.



Automatic Inventory

Tracks items in real time



Expiration Tracking

Reads dates or estimates freshness



Recipe Recommendations

Suggests meals based on what you have



Smart Alerts

Notifies you when items are about to expire

Existing Solutions & Our Differentiation

Smart-fridge concepts already exist. Stockd. focuses on an affordable retrofit approach for everyday households.

SAMSUNG

AI Vision Inside

- ✓ Built-in Refrigerator cameras
- ✓ Recognizes food
- ✓ Smart-fridge ecosystem
- ✗ Expensive, requires new fridge

KitchenPal

- ✓ Pantry manager
- ✓ Meal planner
- ✓ Recipe ideas app
- ✗ Only manual inputs / No hardware

Stockd.

- ✓ Retrofit for existing fridges
- ✓ Automatic IN/OUT tracking
- ✓ Inventory + Freshness
- ✓ Recipe Recommendations
- ✓ Affordable and accessible

Technology

A combination of hardware, AI, and modern web technologies.



Raspberry Pi 5
+ Camera Module



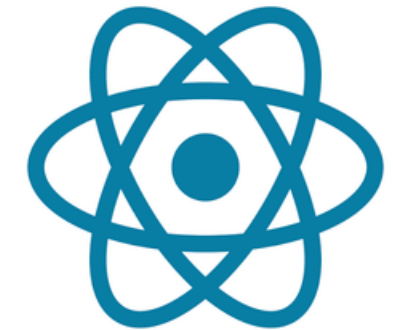
OpenCV
OpenCV
Computer Vision



Python
AI & Backend



FastAPI
AI & Backend



React/Next.js
Web & iPad app

Project Scope

A challenging, but achievable, project for CMSI 4071

Core Development Goals

- ✓ Build responsive app for phones/tablets
- ✓ Create inventory database + backend API
- ✓ Set up Raspberry Pi + fridge camera detection
- ✓ Integrate AI food recognition
- ✓ Sync live inventory and have manual inventory
- ✓ Generate basic recipe recommendation feature

Stretch Goals

- ☐ OCR for expiration dates
- ☐ Weight/quantity sensing
- ☐ Automatic shopping list generation
- ☐ Microphone/voice recognition
- ☐ Barcode scanning

Why This project?



Applies knowledge from previous CS Courses (databases, algorithms, web development)



Gives each team member opportunities to learn new technologies (computer vision, IoT, AI)



Has an appropriate technical challenge



Is realistic to complete in one semester



Uses tools we already have access to



Is interesting, useful, and relevant

Less Waste
More Good Food



Thank you!

Stockd.